XMC7000: Infineon’s new Industrial Microcontroller based on dual Arm® Cortex®-M7

Mar 2023

XMC7000 key features & value proposition

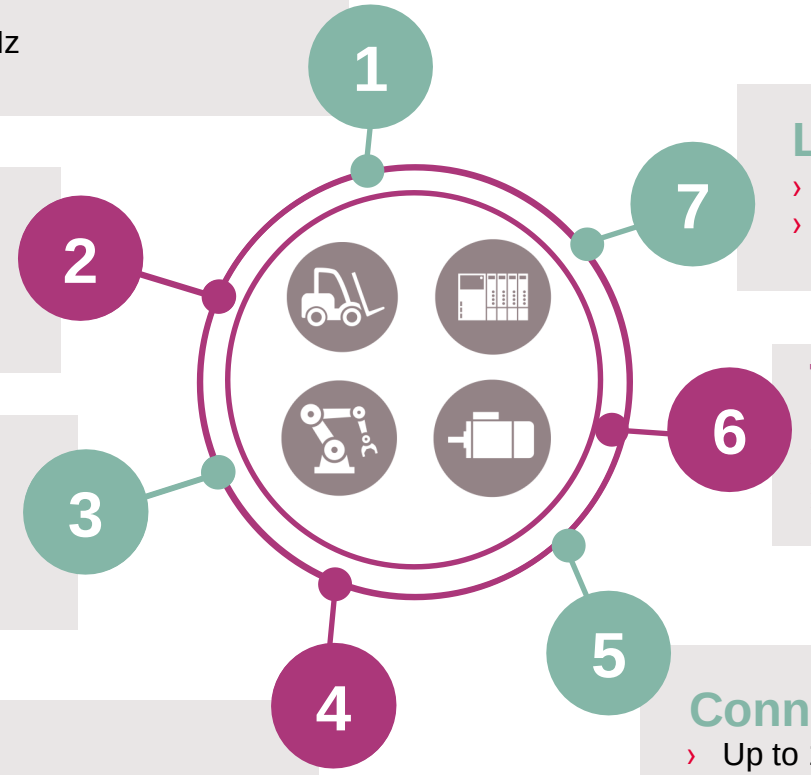
Visit infineon.com/XMC7000 to find out more!

Performance
- Single/Dual Cortex®-M7 Arm® cores, up to 350 MHz
- Cortex®-M0+, 100 MHz
- Up to 8 MB Flash

Updated peripherals
- TCPWM
- SDHC
- Smart I/O

Scalability
- Complete portfolio
- Memory density
- Package lineup

Security
- Secure Hardware Extension (eSHE)
- Hardware Security Module (HSM)



Low power
- Energy-efficient Processing power
- Low power modes

Temperature & Voltage
- Extended temperature range: -40°C up to 125°C
- Extended supply voltage: 2.7 V to 5.5 V

Connectivity
- Up to 10x CAN FD
- Up to 11x SCB
- 2x Ethernet (up to 1 Gbit)

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XMC7000 Extends XMC™ Portfolio

XMC4000-ARM® Cortex®-M4 (with FPU)
› CPU Frequency up to **144 MHz**
› Robust Flash technology: **Ta = -40 to 125°C**
› Timers CCU4, CCU8, Hall & Encoder I/F
› USB / Up to 6x CAN nodes / Up to 6x Serial Channels
› Up to 4x 12 Bit ADC / 2x DAC

XMC4100/4200
Up to 256 kB Flash / 40 kB RAM
48-64 pins

XMC4400
Up to 512 kB Flash / 80 kB RAM
64-100 pins
› 120 MHz Core
› Ethernet
› ΔΣ Demodulator

XMC4500
Up to 1 MB Flash / 160 kB RAM
100 – 144 pins
› EBU
› SD Card

XMC4700
Up to 2 MB Flash / 352 kB RAM
100 – 196 pins
› 144 MHz Core
› 6 ch CAN

XMC4800
Up to 2 MB Flash / 352 kB RAM
100 – 196 pins
EtherCAT®

XMC4300
256 kB Flash / 352 kB RAM
100 pin

XMC1000-ARM® Cortex®-M0
› Core up to **48 MHz** / Peripherals up to **96 MHz**
› Timers CCU4,CCU8
› 2x Serial Channels
› 12 Bit ADC
› **1.8 V-5.5 V**
› Ta = -40 to 105°C

XMC1100
up to 64 kB Flash
16 – 40 pins
› 9 ch LED flicker free control (BCCU)
› 3x Analog Comparators

XMC1200
up to 200 kB Flash
16 – 40 pins
› Math Co-Processor
› CCU8 PWM Timer
› Hall & Encoder I/F

XMC1300
up to 200 kB Flash
16 – 40 pins
› 48 MHz/96 MHz clock
› 2x CAN
› 2x CCU8
› 4x Analog Comparators

>70% performance increase

XMC1400
up to 200 kB Flash
40 – 64 pins

XMC7100
4 MB Flash / 768 kB RAM,
250 MHz single/dual core
100 – 272 pin (QFP/BGA)

XMC7200
8 MB Flash / 1 MB RAM,
350 MHz single/dual core
176 – 272 pin (QFP/BGA)

› **Single** or **dual core** Arm® Cortex®-M7 and Cortex®-M0+
› M7 CPU core Frequency up to **350 MHz**
› Temperature range: **Ta = -40 to 125°C**
› Memory Up to **8 MB Flash, 1 MB RAM**
› **TCPWM** timers of 102Ch 16 bit and 16Ch 32 bit
› **Cryptography** Engine support
› Interfaces such as CAN-FD, **Gb Ethernet**
› **SMIF** and SDHC interface supported
› Extended voltage operating range: **2.7 to 5.5V**
› low power modes(LP/deep sleep, **hibernate**..)
› Add. timer, ADC and Serial COM. Chs
› **Read While Write (RWW)** Flash capability

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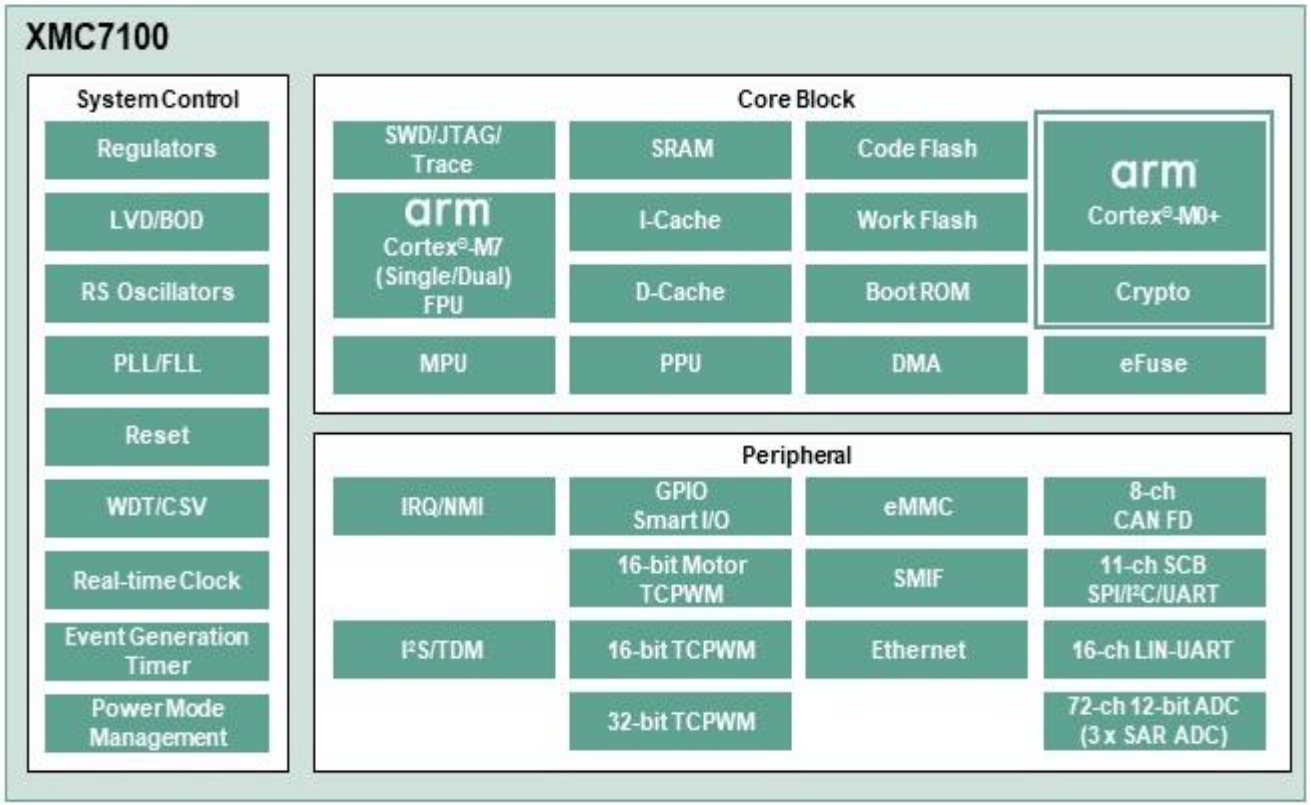
XMC7100 series



Description

Applications / Target Markets

› Industrial motor control and drives, inverters/converters (UPS, solar), High-end eBikes, LEVs, eV Chargers, PLCs

Feature highlights

› **32-bit MCU Core System**
  • Single/Dual 250-MHz Arm® Cortex®-M7 and 100MHz Cortex®-M0+
  • Up to 4-MB Flash, up to 768-kB SRAM, I/D-Cache
› **2.7-V to 5.5-V** Supply Voltage
› **Up to 125°C** extended temperature range
› **Interfaces:**
  • Up to 8-ch CAN FD, up to 11-ch SCB
  • eMMC, SMIF (QSPI/HS-SPI), 1-ch 10/100 Mbit Ethernet
› **AD Converter**
  • Up to 72-ch, 12-bit with 3x successive approximation ADC (SAR ADC) units
› **Timers**
  • Up to 12-ch 16-bit motor control, 63-ch 16-bit timer/counter/pulse-width modulation (TCPWM), and 8-ch 32-bit TCPWM
  • Event Generation Timer

Packages

QFP-100
QFP-144
QFP-176

BGA-272

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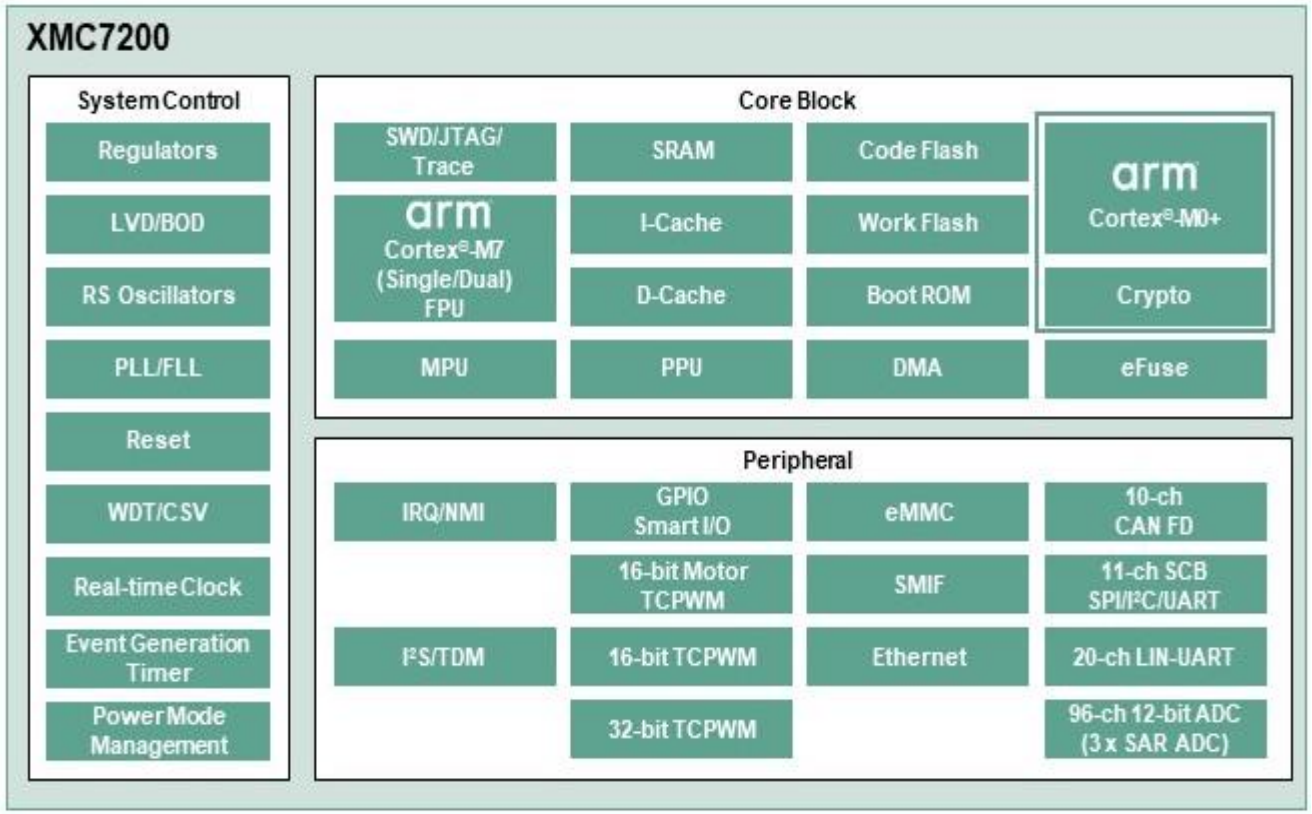
XMC7200 series



Description

Applications / Target Markets

› Industrial motor control and drives, inverters/converters (UPS, solar), LEVs, eV Chargers, PLCs

Feature highlights

› **32-bit MCU Core System**
  - Single/Dual 350-MHz Arm® Cortex®-M7 and Cortex®-M0+
  - Up to 8-MB Flash, up to 1MB SRAM, I/D-Cache

› **2.7-V to 5.5-V** Supply Voltage

› **Up to 125°C** extended temperature range

› **Interfaces:**
  - Up to 10-ch CAN FD, up to 11-ch SCB
  - eMMC, SMIF (QSPI/HS-SPI), up to 2-ch 10/100/1000 Mbit Ethernet

› **AD Converter**
  - Up to 96-ch, 12-bit with 3x successive approximation ADC (SAR ADC) units

› **Timers**
  - Up to 15-ch 16-bit for motor control, 87-ch 16-bit timer/counter/pulse-width modulation (TCPWM) and 16-ch 32-bit TCPWM
  - Event Generation Timer

Packages

QFP-176

BGA-272

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Key features and benefits

Infineon's XMC7000 at a glance: high quality grade industrial MCU platform

Key features	Key benefits	Value
Single/Dual Cortex®-M7 Arm® cores, up to 350 MHz and memory up to 8 MB	› Perfect fit for demanding industrial application use cases › Higher processing performance › Ability of task distribution	Best-in-class compute performance
Comprehensive set of advanced peripherals (ADCs, TCPWM Timers) along with competitive set of security offering	› Optimal solution for motor control & power conversion applications › Advanced security options	Allowing developers to create feature optimized end products
High quality and temperature grade of 125°C in combination with lower-power modes down to 8 µA	› Low power to support electrification and digitization › Power saving in energy-critical applications › Ability to operate in high ambient environments	Increased flexibility and quality
Four footprint versions available in 17 different part numbers	› XMC7000 is easy to adapt on existing PCB › Different core/memory/package combinations	Optimal device arrangement

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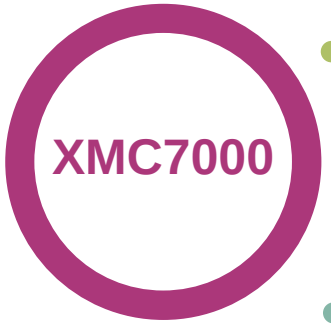
Target applications and key features



Industrial drives & Robotics

- Up to 350MHz cloc for low/mid-end industrial drives
- TCPWM: standard 16-bit timers, dedicated to motor control 16-bit counters and 32-bit timers
- Extended voltage range: 2.7-5.5V
- Up to 125°C extended temperature range

EV-charging

- Large number of specialized TCPWM timers and ADCs
- HW security support
- Up to 125°C extended temperature range
- Interfaces: 10-ch CAN FD and 16-ch UART

Electric 2-wheeler

- TCPWM timers, standard (16-bit) and motor control timer (32-bit)
- Interfaces: 10-ch CAN FD and 16-ch UART
- Energy efficient processing power: different power savings mode

PLC

- Dual 350-MHz: more robust (redundant) device performance for distributing the tasks
- 10/100/1000 Mbit Ethernet interface
- Up to 8MB high memory footprint
- 12-bit ADCs with sampling rates of up to 1 Msps

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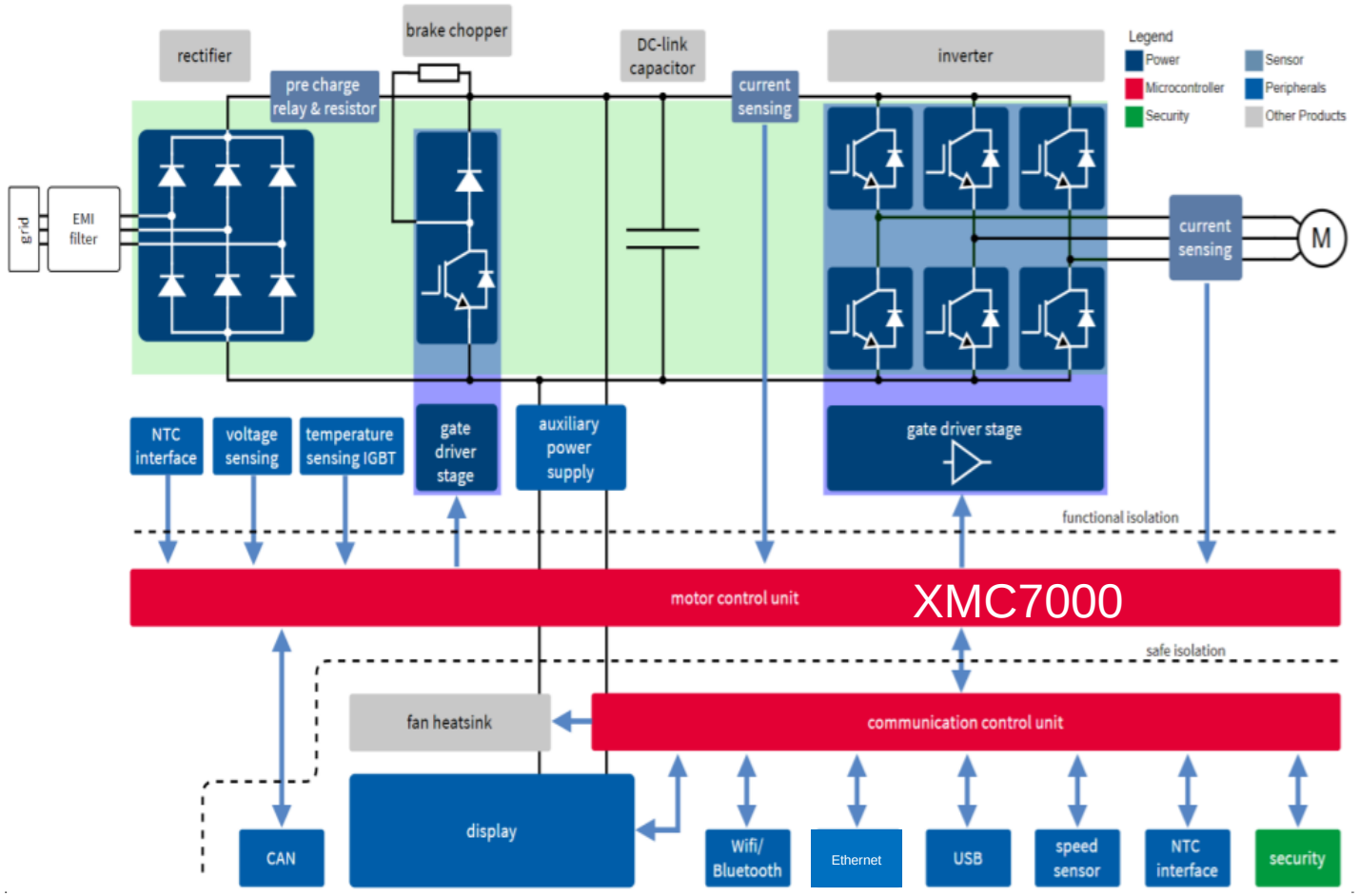
XMC7000- Industrial Drives

Application block diagram



Key features

- 250-MHz clock speed suited for low-end industrial drives
- Dual core Arm Cortex M7 with Arm Cortex M0
- TCPWM timers(118Ch), 15-Ch 16bit for motor control , 16Ch 32 bit as seperate TCPWM group.
- 3 ADC covertor with 99 channels (3 dedicated for motor control)
- 10/100/1000 Mbps Ethernet MAC interface
- Cryptography engine – HW security
- Extended Voltage range: 2.7-5.5V
- Up to 125°C extended temperature range

Target Applications

- Servo Drives
- GPD –General purpose drives (compact, standard,Premium)

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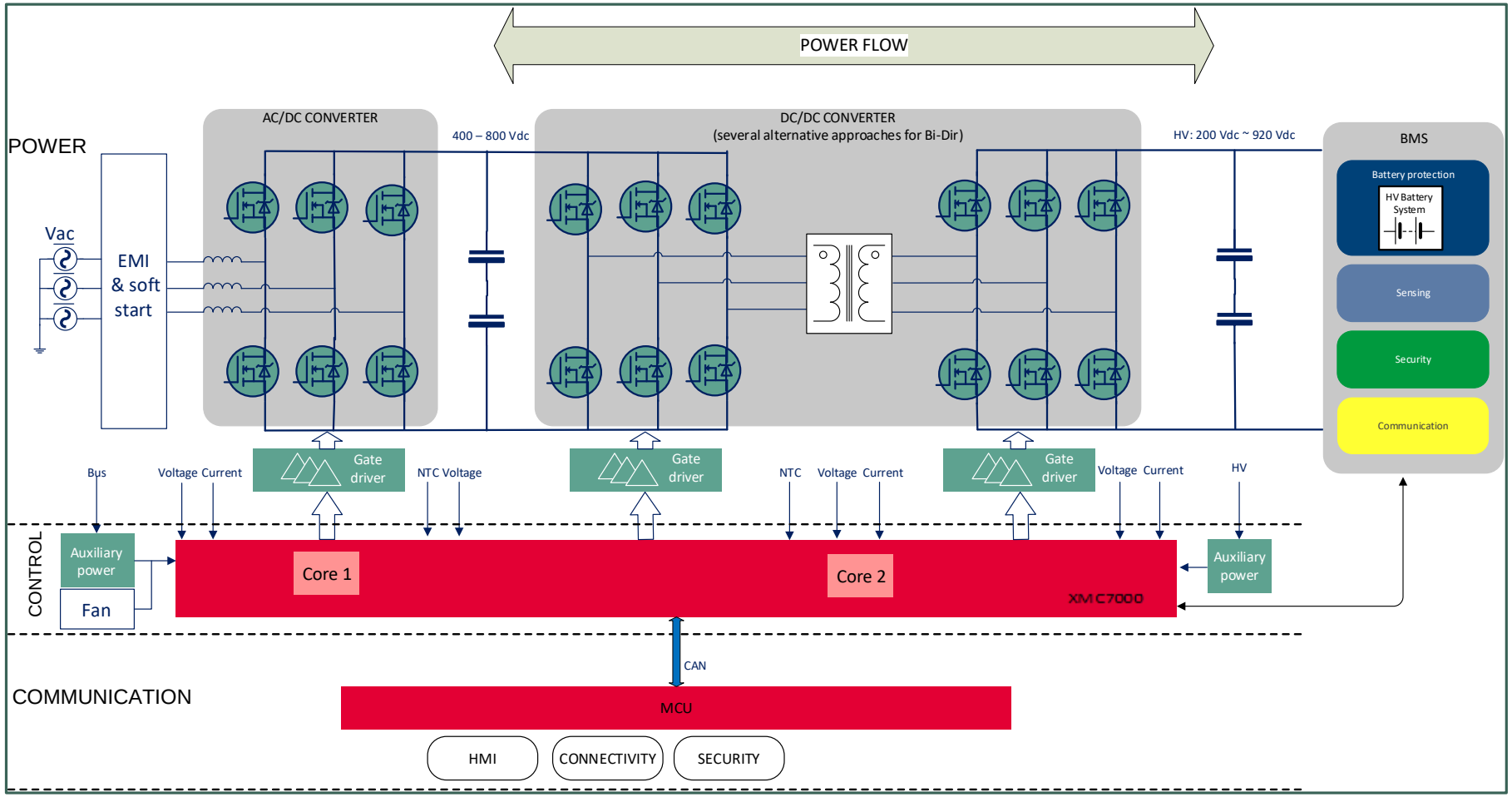
XMC7000- EV Charging

Application block diagram



Key features

- 10/100/1000 Mbit Ethernet
- HW security support
- Up to 125°C extended temperature range
- Interfaces: 10-ch CAN FD and 16-ch UART
- Dual 350-MHz: more robust device performance for task distribution
- TCPWM timers
- Large numbers of ADC channels

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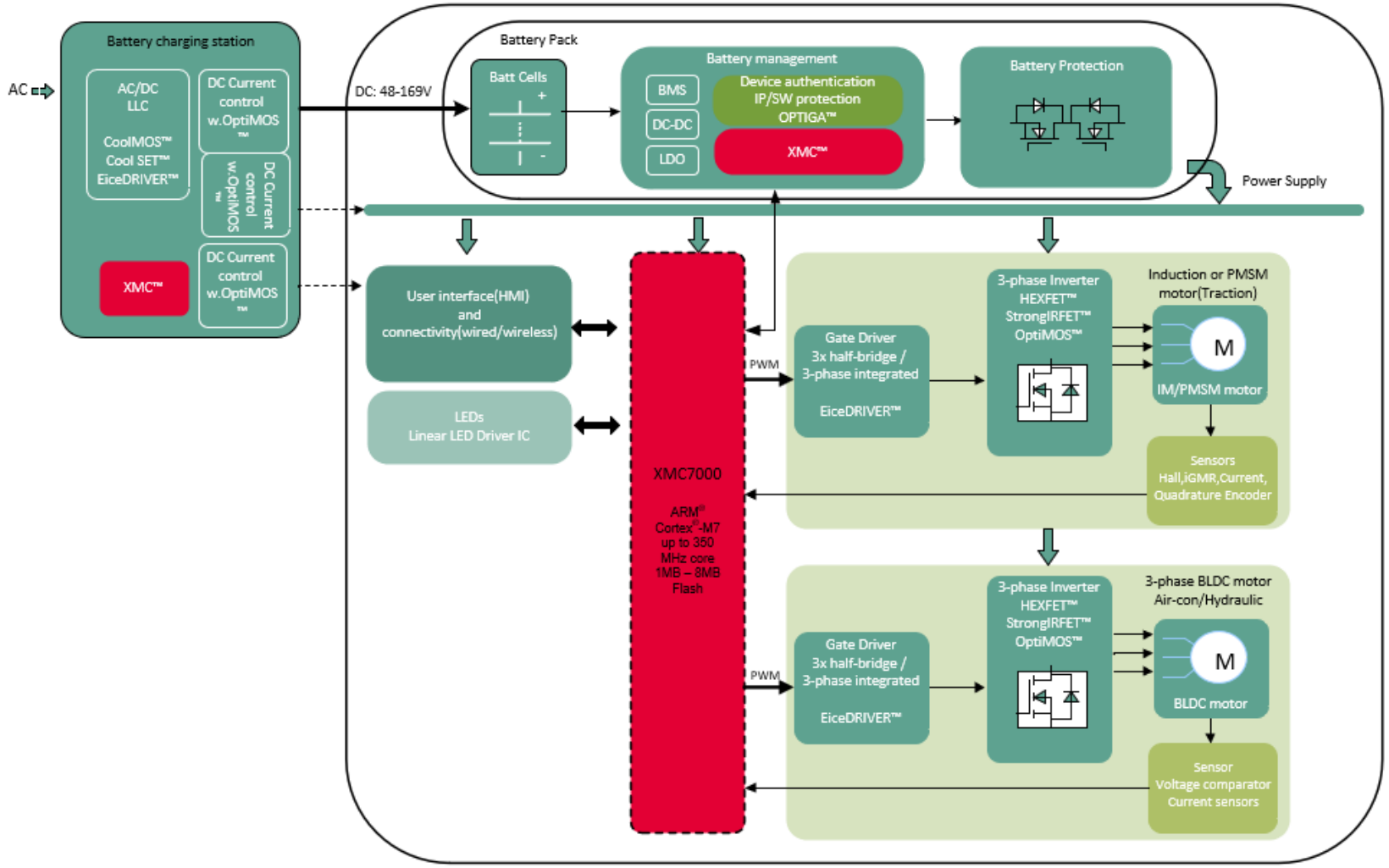
XMC7000- LEV (Light Electric Vehicle)
High power low speed/Low power light electric vehicle

Application block diagram



Key Features

- Dual core Arm Cortex M7 with Arm Cortex M0+ for cryptographic engine
- TCPWM: standard 16-bit timers, motor control dedicated 16-bit counters and 32-bit timers
- Timer supporting Quadrature decoding
- Interfaces: 10-ch CAN FD and 16-ch UART
- 5 Smart I/O block to perform Boolean operation
- Energy efficient processing power: different power savings mode

Target Applications

- E-forklifts
- E-golf carts
- Low speed E-vehicles(LSEVs/micro Ev)
- E-bikes
- E-scooter (standing, self balancing etc)
- E-Rickshaws or E-three wheeler

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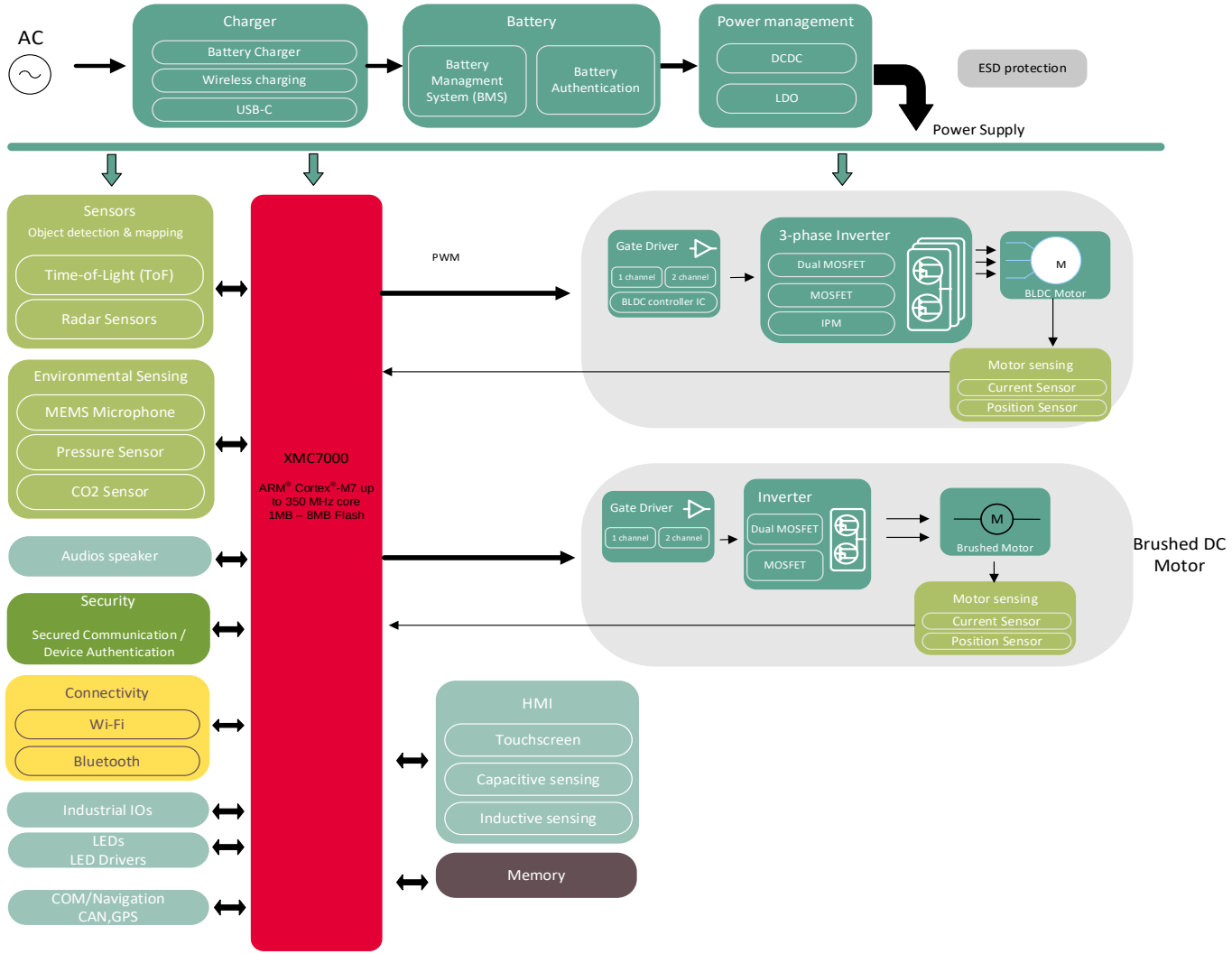
XMC7000- Industrial and Residential Robotics

Application block diagram



Key Features

- Dual 350-MHz: more robust device performance for task distribution
- Up to 8MB high Flash memory footprint and 1024 KB SRAM
- 3x 12-bit ADCs with sampling rates of up to 1 MSPS
- Flexibile and powerful TCPWM timers
- External memory - Single , dual, quad, octal SPI or HYPERBUS interface
- Up to 125°C ext Temperature range

Target Applications

- Mobile robots(AGV,AMR)
- Industrial robots
- Service robots
- Multicopters and drones
- Cobots (colloborative robots)

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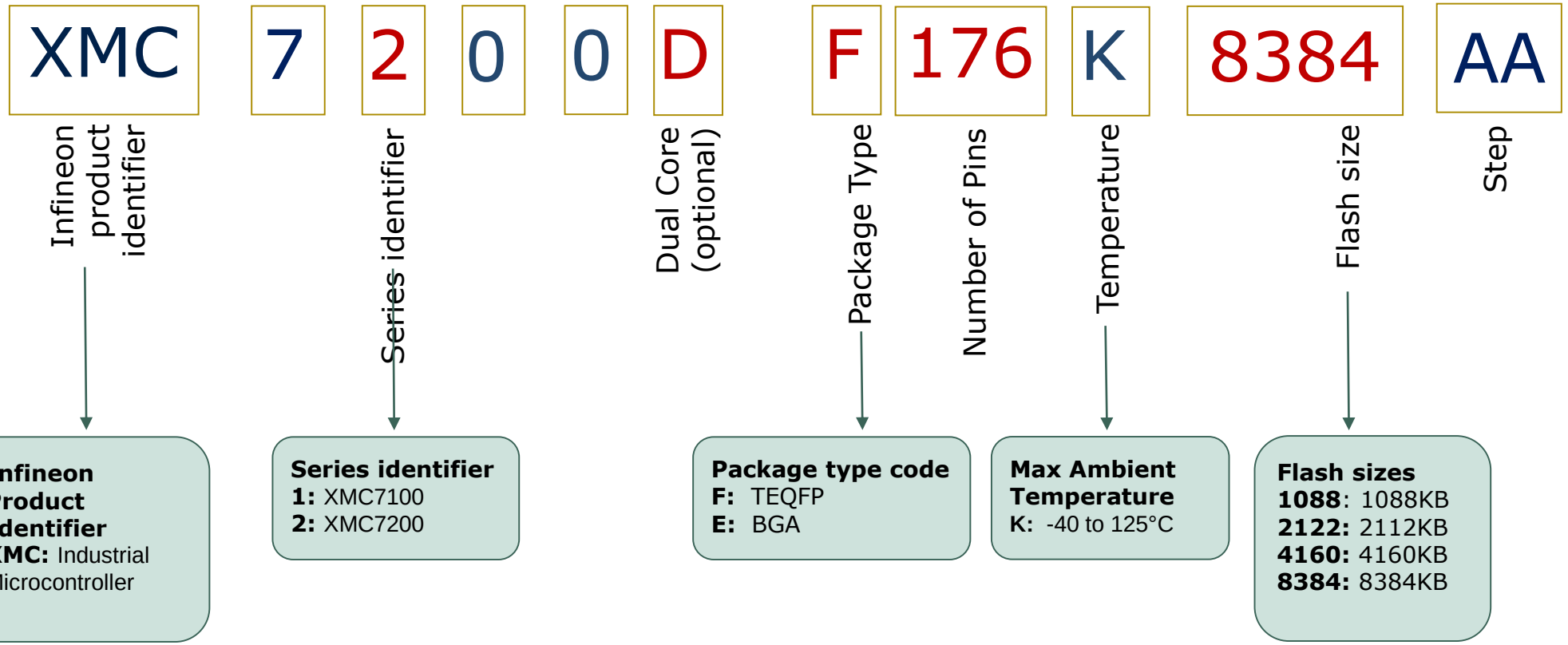
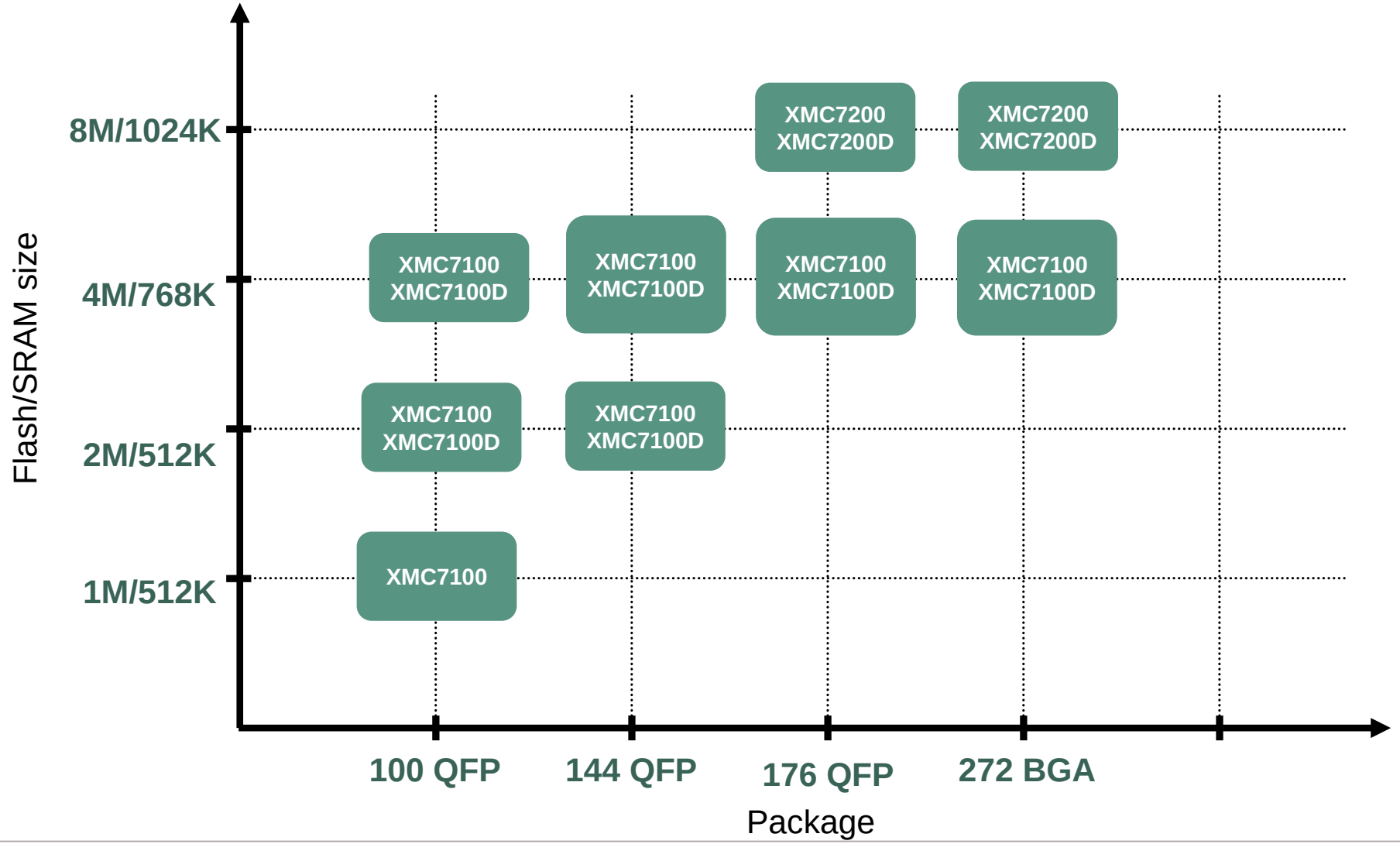
XMC7000 Part number coding



Available part numbers
XMC7100-F100K1088AA
XMC7100-F100K2112AA
XMC7100-F100K4160AA
XMC7100D-F100K2112AA
XMC7100D-F100K4160AA
XMC7100-F144K2112AA
XMC7100-F144K4160AA
XMC7100D-F144K2112AA
XMC7100D-F144K4160AA
XMC7100-F176K4160AA
XMC7100D-F176K4160AA
XMC7100-E272K4160AA
XMC7100D-E272K4160AA
XMC7200-F176K8384AA
XMC7200D-F176K8384AA
XMC7200-E272K8384AA
XMC7200D-E272K8384AA

XMC7000 portfolio matrix



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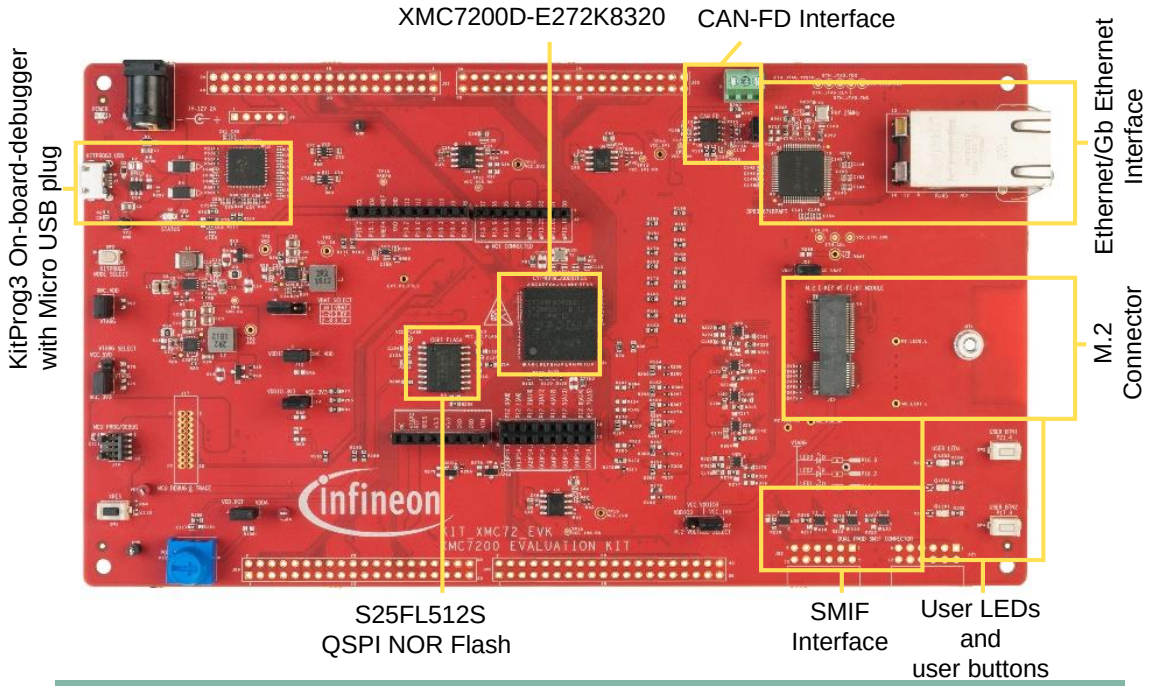
The XMC7200 Evaluation Kit: quick testing and prototyping

XMC7200 Evaluation Kit



Applications

Features

- Evaluation board for XMC7200D-E272K8384 dual-core ARM® Cortex-M7 device running up to 350MHz
- Full system approach on the board, featuring Gigabit Ethernet PHY and connector, CAN-FD transceiver, NOR flash memories, user LEDs, buttons and potentiometers
- Modular system enablement through Arduino and M.2 connectors

Customer Benefits

- Out-of-the box experience with on-board-debugger and DC power supply provided
- Full integration in Modus Toolbox IDE through Board Support Package (BSP)
- Easy access and evaluation of all peripherals using code examples available in BSP
- Arduino and M.2 ecosystem access

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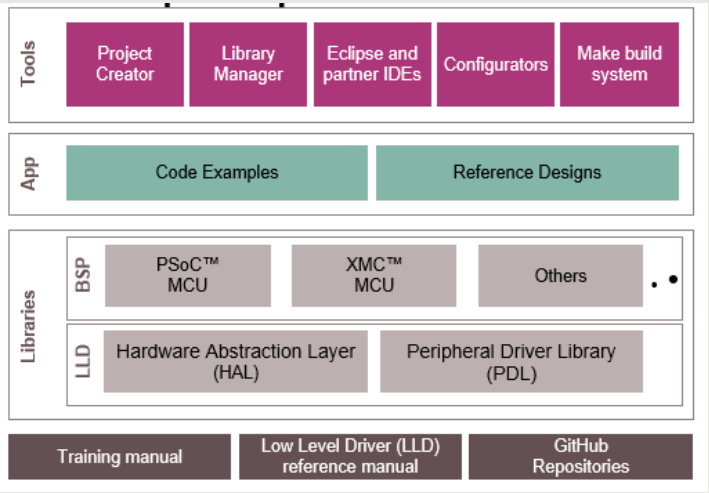
XMC7000 in ModusToolbox™ Software – Overview



- ModusToolbox™ software is a modern, extensible development environment supporting a wide range of Infineon microcontroller devices.
- Provided as collection of development tools, libraries, and embedded runtime assets architected to provide a flexible and comprehensive development experience.

Development Tools

The ModusToolbox™ tools package includes desktop programs that enable the creation of new embedded applications, managing software components, configuring device peripherals and middleware and embedded development tools for compiling, programming, and debugging.

Run-Time Software

The ModusToolbox™ software includes an extensive collection of GitHub-hosted repositories comprised of code examples, board support packages, middleware, and application support.

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